

BrickAtlas: Measuring Visual, Structural, and Executable Reasoning on Shared Brick Assemblies

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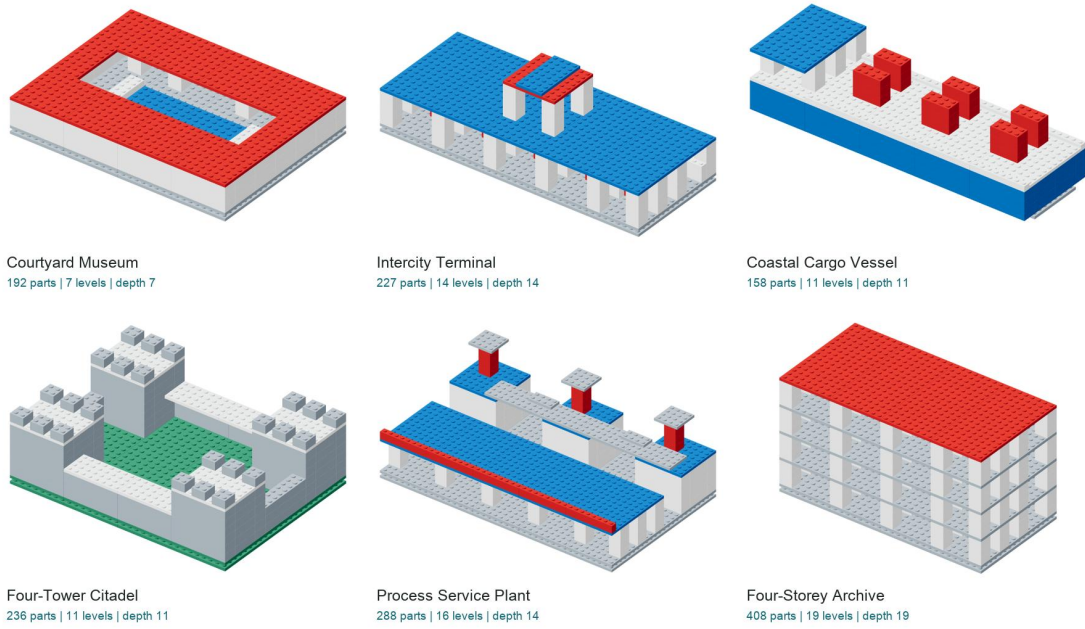


Figure 1. **All six new Frontier structures.** Original 158–408-part models add deep courtyards, service galleries, hollow towers and stacked floors. Eight contracts per object connect maintenance, access certificates, resource constraints and uncertainty. Diagnostic-2 uses these sources for matched-context maintenance, layer-reference perception and geometric inspection. New learned-model results remain unmeasured; the historical population is separate.

Abstract

Understanding a brick assembly requires distinguishing its appearance, hidden structure, requested changes, and executable construction. BrickAtlas separates appearance, structural change and executable success with explicit input privileges. Its historical audit contains 5,120 procedural sources and 117,910 conditions. A new Frontier release adds six original 158–408-part objects. Diagnostic-2 supplies 138 paired conditions and independently elicitable diagnosis, access and repair stages. Constant-answer audits uncover and replace two legacy task shortcuts; matched contexts preserve exactly the same answer. New learned-model results remain unmeasured. A finite-grammar ambiguity track supplies 600 paired conditions with exhaustively

checked answer sets, and 90 exterior-only queries are genuinely undetermined within the declared grammar. A separate renderer audit verifies identical exterior pixels across admissible layouts. An eight-model development matrix and same-source algorithmic controls expose different failure patterns, while full-casebank copying achieves 87.1% completion F1 and 97.4% removal F1 without completing either task. We release replayable outputs and source-level audits. The model sample is small; independent confirmation, richer licensed objects, and real human metric calibration remain outstanding.

1. Introduction

A system can recognize the bricks in an object without recovering their poses, match its exterior without identifying hidden connections, or return a nearly unchanged structure with high similarity after failing an edit. A correct final assembly also need not make a submitted construction order executable. Treating these outcomes as a single notion of “structural intelligence” obscures what a model accomplished.

BrickAtlas uses shared source assemblies to separate these endpoints. Figure 3 shows actual inputs and ground truth, while Fig. 5 distinguishes a valid sequence from a dead end. Following dataset/model reporting and holistic evaluation principles [5, 15, 22], we separate information, outputs, equivalences and efficiency.

This is not a claim to originate brick generation, reconstruction, or multi-model assembly evaluation. BrickGPT [25] and BrickNet [14] provide strong generative foundations; MEPNet [31], Neural Assembler [32], and TreeSBA [8] address visual assembly. BC-Bench [16] evaluates selection and pose; PhyBlock [18] covers diagnosis and planning robustness. Our narrower opportunity is *measurement validity across shared objects*: when does a plausible appearance, a high similarity score, or one valid reference justify a downstream structural conclusion?

Our central question is whether individually plausible answers encode structural knowledge consistent with downstream execution. Diagnostic-2 operationalizes diagnosis–access–repair coherence separately from correctness, with matched local/full contexts and symbolic/layer references. This new model experiment is unrun. Historical studies examine three supporting questions: global similarity versus requested changes, hidden-relation identifiability, and candidate reordering beyond identical-repeat instability.

The release contributes (1) a source-grouped eight-family evaluation contract with visual, structural, and executable endpoints; (2) a new finite-grammar, observation-conditioned relation track with exhaustive ground-truth sets and public-input enumeration; and (3) full-population short-cut audits, an eight-model descriptive matrix, and paired protocol controls; and (4) a versioned challenge layer for maintenance certificates, minimum intervention, resource constraints and finite-world inspection. We separate implemented capability, measured development evidence, and unrun experiments throughout.

2. Related Work and Position

Generation and assembly. BrickGPT [25] generates stable and buildable objects from text. BrECS [1] studies budget-aware generation and completion with efficient constraint checking. BrickNet [14] uses human-designed LDraw objects and typed connectors; its 320,808 sam-

ples and 9,743 part variants far exceed our grid vocabulary. Nominal contact in BrickAtlas is not a substitute for their structural or physical scope. MEPNet [31] reads stepwise manuals; Neural Assembler [32] predicts components, topology, poses, and sequence from multiple views; TreeSBA [8] uses a tree representation and synthetic-to-real supervision. Break and Make [30] studies interactive inspection, disassembly, and rebuilding. Their input protocols differ from fixed image or symbolic submissions and require audited adapters for comparison.

The closest multimodal benchmark. BC-Bench/Brick-Composer [16] evaluates diverse-brick selection and pose estimation from manual context and current state, including feedback and training. LEGO-Puzzles [29] also includes multi-step plans; PhyBlock [18] studies progressive planning, diagnosis and counterfactuals. Co-builder [10] addresses state detection, and WorkBenchMark [19] includes robot/VLA assembly. These are not empty research areas. Our bounded distinction is shared-object, costed maintenance and observation-consistent certificates, not first-ever assembly reasoning.

Diagnostic visual reasoning. CLEVR [13] establishes controlled compositional reasoning evaluation; BLINK [4] tests perceptual skills, including multi-view and spatial reasoning. CSGNet [28] predicts executable shape programs. SpatialBabel [17] examines reconstruction across scene-code languages. Output-protocol effects are therefore not new in themselves. Answer consistency also has prior work [11, 12]; our distinction concerns executable maintenance constraints. Similarly, MCQ position bias is established [33]; our candidate experiment specifically compares actual pose changes against an identical-repeat baseline, without inferring internal model mechanisms from external API replies.

3. Benchmark Construction and Tasks

3.1. Geometry, Sources, and Splits

A part is (i, t, c, x, y, z, r) : instance ID, type, color, minimum body corner, and quarter-turn yaw. **Y is vertical**; X, Z use 8 mm studs and Y uses 3.2 mm plate heights. Origins are integer coordinates in $[0, 24]^3$, bodies must fit in $[0, 32]^3$, and submissions contain at most 64 parts. Odd yaw swaps width and depth. Geometry is nominally valid if bodies do not overlap and every elevated part has at least one directly supporting stud. Force balance, clutch, arbitrary CAD geometry, and robotic grasping are outside this contract.

Eight seeded policies accept 640 unique sources each, all starting from a 4-by-8 plate. Canonicalization removes color and translation and considers four yaw rotations, but

Work	Supplied information	Main output / setting	BrickAtlas distinction
BrickGPT; BrECS	Text or target constraints	Generation / completion	Multi-task measurement, not new generation
BrickNet	LDRAW parts and typed connectors	Graph-backed build programs	Smaller domain; explicit answer equivalence
MEPNet; TreeSBA	Manuals or multi-view images	Poses / assembly actions	Shared task labels and changed-region metrics
Break and Make	Interactive observations	Disassembly and rebuilding	Fixed-input and tool tracks kept separate
BC-Bench	Manual, current state, candidate bricks	Selection and single-step pose	Hidden-layout ambiguity and typed multi-task success
SpatialBabel	Images and scene-code languages	Primitive scenes / spatial QA	Nominal contacts, edits, and original-order execution

Table 1. Closest-work positioning by input and output protocol. This is not a capability checklist or an empirical comparison on matched data. Specialized-method adaptation remains unrun.

Split	Sources	Task conditions
train	2,690	61,870
validation	392	9,016
test_id	758	17,584
test_ood	1,280	29,440
Total	5,120	117,910

Table 2. Source-group split. ID cases additionally include 150 isolated recognition questions, not 150 new assembly objects.

not mirrors; within-release and earlier-release exact duplicates are rejected.

Six policies use a deterministic geometry-hash split, while bridge and shell are policy-held-out (Table 2). All task variants and prefixes retain the source group. This is a procedural-policy holdout, not absence of shared primitives or a semantic generalization guarantee. An exhaustive 8,305,836-pair cross-split audit finds no canonical-token Jaccard similarity at least 0.8; it does not test best-aligned subassemblies or pretraining contamination.

Historical sources contain 9–47 parts and 19 assembled types from the 25-type catalog. Policy and size are confounded; the supplement gives the full census. More views do not create independent objects.

Curated and challenge layers. The twelve 8–28-part curated objects and six 49–63-part Challenge-2 objects remain calibration layers. Frontier-1 (Fig. 1) adds six 158–408-part objects, 1,509 placed parts, 14–24-stud depth, and dependency chains up to 19. Its separate 512-part, 96-cell envelope does not change historical limits. Eight tasks per object cover minimum-action service, access certificates, parallel scheduling, minimum support intervention, inventory replacement, hidden seam sets, contingent inspection and mixed-fault repair. An audit found identical answers across

all six legacy ambiguity and inspection cases. Diagnostic-2 replaces these templates and adds 72 paired maintenance conditions, 36 geometry-observation conditions, 18 deadline schedules and 12 mixed-pose conditions. Full references are withheld from the layer-image track. The supplement defines all new contracts.

3.2. Task and Information Contract

Table 3 defines the task families. **Recognition** predicts a known catalog type, color, and stud count from isolated views. **Relations** uses a symbolic assembly and two IDs to ask direct contact, vertical ordering, mating-stud count, and shortest undirected contact-graph distance; a non-contact pair can still have a graph path. **Reconstruction** predicts a full structure from reference information and BOM. **Generation** accepts any legal structure satisfying exact extents, zero anchoring, a piece budget, color count, connected component count, and minimum maximum contact degree. It is constrained layout generation, not semantic text-to-shape evaluation.

Completion removes the last $\lfloor n/4 \rfloor$ pieces of a valid insertion sequence and requires preserving the remaining prefix. It does not cover arbitrary holes. **Editing** rotates the whole structure, replaces all occurrences of one color, or removes one topmost piece. Rotation maps positive X toward positive Z , keeps Y , and reanchors minimum X, Z . **Repair** has no-fault, color-fault, and one-stud-shift cases on exposed topmost pieces. It requires corrected geometry and faulty current IDs, with an empty fault list for a normal input. **Planning** supplies the target with IDs in lexical order and requests any executable permutation for vertical assembly/disassembly.

Each source yields 13 variants and 23 information-conditioned cases. Reconstruction, completion, and all three repairs have three reference conditions. With 25 catalog types in six colors, the total is $5,120 \times 23 + 150 =$

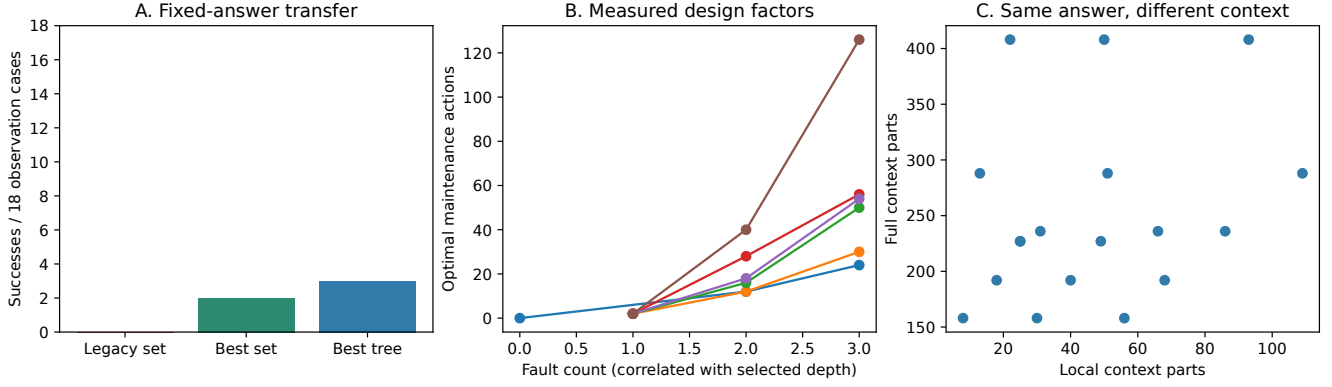


Figure 2. Diagnostic-2 design audits, not learned-model performance. Fixed-answer transfer uses shared candidate IDs; the best released set and tree transfer to 2/18 and 3/18 cases. Matched contexts have identical answers despite different total sizes. Fault count and selected depth remain correlated; these are not fully orthogonal difficulty strata.

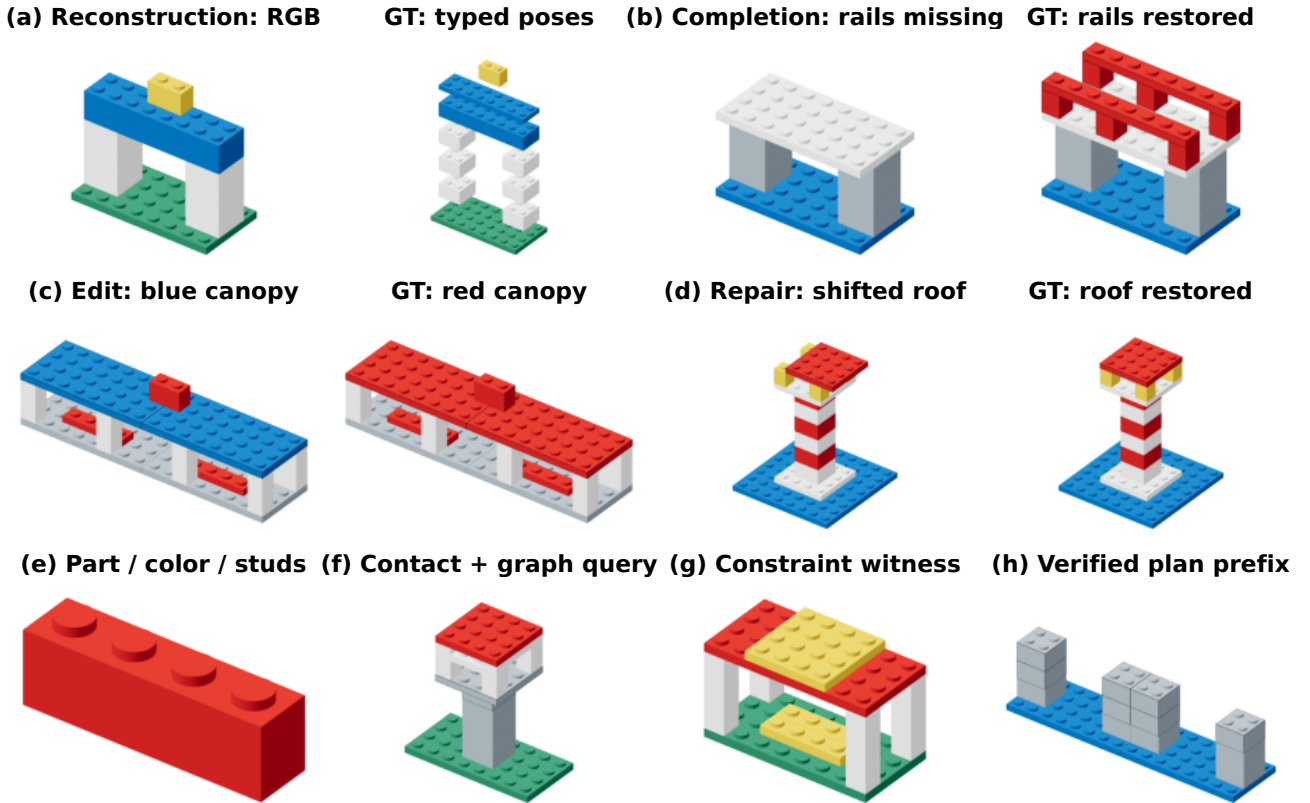


Figure 3. **Task-specific curated input and ground truth.** Reconstruction maps gate views to typed poses; bridge completion restores rails; station editing changes only the canopy color; lighthouse repair restores the shifted roof. Bottom panels illustrate part recognition, a typed relation query, one valid generation witness, and a verified planning prefix. Exploded offsets are display-only, not valid poses or ordinary model observations.

117,910. **Ordinary RGB** provides isometric/top/front/side assembled views and BOM; interiors remain hidden. **Layer disclosure** provides an assembled view and top images of each bottom- Y layer with other layers removed. **Symbolic**

reference supplies complete geometry. The latter two add information and require full type/color/pose recovery. Other families use their declared symbolic or constraint inputs. Fault labels, changed IDs, and private render recipes are ex-

Family (variants)	Public input	Output	Success contract	Cases
Part recognition	2 isolated RGB views	Type, color, stud count	All three fields correct	150
Relations (2)	Symbolic structure + two IDs	Connection, above, studs, distance	All four fields correct	10,240
Reconstruction	Reference + BOM	Full structure	Condition-specific match + legality	15,360
Generation	Six public constraints	Any satisfying structure	All constraints + legality	5,120
Completion	Current prefix + reference + BOM	Full structure	Match + preserved prefix	15,360
Editing (3)	Current structure + operation	Full edited structure	Exact edit + protected preservation	15,360
Planning (2)	Target structure + direction	Permutation of all IDs	Complete executable order	10,240
Repair (3)	Current + reference + BOM	Structure + faulty current IDs	Match + preservation + localization	46,080

Table 3. Eight task families. “Reference” denotes the declared RGB, layer, or symbolic condition. Symbolic tasks are not visual scores.

cluded from ordinary prompts.

4. Ground Truth and Measurement

4.1. Visual Compatibility Versus Exact Structure

Ground-truth records bind input hashes, source groups, target/source geometry, protected and changed IDs, contacts, nominal stud mates, executable witnesses, and metric applicability. Multiset matching ignores arbitrary output IDs and symmetric per-part yaw, but does not silently align global translation or rotation. For m matched parts in target T and answer A ,

$$P = m/|A|, \quad R = m/|T|, \quad F_1 = 2m/(|A| + |T|). \quad (1)$$

BOM F1 removes pose; occupancy IoU removes part boundaries; edge F1 requires matched endpoints and mating-stud counts.

For ordinary reconstruction, let S^+ be colored first-hit cells along positive X, Y, Z . We accept

$$V(A, T) = \mathbf{1}\{\text{valid}(A)\} \mathbf{1}\{\text{BOM}(A) = \text{BOM}(T)\} \cdot \mathbf{1}\{S^+(A) = S^+(T)\}. \quad (2)$$

This proxy allows alternative hidden arrangements; it is not arbitrary camera visibility or pixel similarity. Full-reference correctness is reported separately. Completion subtracts preserved source matches before computing *addition* F1. Edits/repairs require protected piece preservation; repairs additionally require faulty-ID localization. False alarms are defined only for no-fault cases.

RQ1: no-operation shortcuts. Copying a prefix with k missing parts from an n -part target gives

$$F_1^{\text{copy}} = \frac{2(n-k)}{2n-k}, \quad F_{1,\text{added}}^{\text{copy}} = 0. \quad (3)$$

Copying before a one-piece removal similarly yields $2(n-1)/(2n-1)$, approaching one without performing the edit. A full 35,840-case no-operation census confirms this: 87.1% completion and 97.4% removal F1, both with zero task success. Positive repair copies reach 94.9% F1 without correct localization. Normal no-fault repair correctly succeeds unchanged. All seven endpoints contain one case per source; per-case results and complete global/local metrics remain available.

4.2. An Ambiguity-Aware Extension

Matching appearance need not identify hidden relations. Instead of forcing one reference answer, we make the admissible set explicit. The new, separately versioned track uses sealed shells with a $2 \times W$ internal layer, $W \in \{2, 4, 6\}$, tiled by identical 1-by-2 bricks. Three stack heights give nine parameterized families. The tilings are exhaustive within this finite grammar: their counts are 2, 5, and 13, independently checked by the recurrence $c_W = c_{W-1} + c_{W-2}$, $c_0 = c_1 = 1$. There are 60 valid layout states across the nine families.

For each pair of blue upper bricks we ask whether they share a directly supporting red brick. Given the observation O and public grammar $\mathcal{G}$, let $\mathcal{A}(O, \mathcal{G})$ be all admissible layouts. With Boolean query q ,

$$y(O) = \begin{cases} \text{yes}, & \forall A \in \mathcal{A}, q(A) = 1, \\ \text{no}, & \forall A \in \mathcal{A}, q(A) = 0, \\ \text{undetermined}, & \text{otherwise.} \end{cases} \quad (4)$$

Exterior and support-disclosed conditions create 600 paired cases (Table 4). Ninety exterior questions are genuinely ambiguous within the declared grammar; disclosure resolves them. Known geometry and the grammar are supplied symbolically, so this extension tests observation-consistent structural inference, not pure visual perception. Public

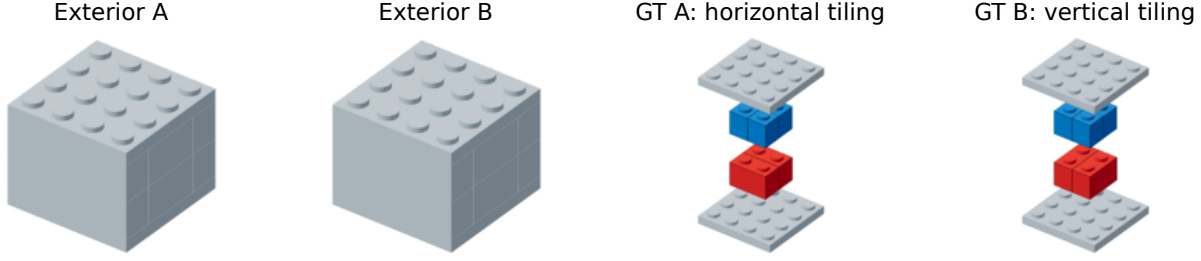


Figure 4. **Same exterior, different ground truth.** The two sealed structures have identical rendered exterior pixels under the tested cameras. Exploding their layers reveals horizontal versus vertical red internal bricks; side walls are omitted in the disclosed views. The outlined adjacent blue pair shares one red supporter in A but not B. The exterior-only answer is therefore *undetermined*; disclosing the support layer makes the answer definite. Exploded views are evaluator illustrations.

Condition	Groups	Cases	Yes	No	Undet.
Exterior	9	300	0	210	90
Support disclosed	9	300	36	264	0

Table 4. New finite-grammar task counts, separate from v2. Nine parameterized families are not nine independent semantic sources.

records omit labels and hidden supports; an enumeration solver consumes only these public inputs and solves all 600 cases.

All layouts agree in BOM, colored occupancy, and declared surfaces and pass two geometry implementations. A separate high-resolution renderer checks all 60 layouts in four views: 240 rendered frames match their within-family reference pixel-for-pixel. This result is renderer/camera-specific, not human calibration. Unweighted always-no accuracy is 70% exterior and 88% disclosed; class-balanced accuracy is 50% in both. We therefore report balanced success, false certainty on ambiguous cases, and unnecessary abstention on determinate cases. Complete baseline and planned VLM tables are in the supplement.

4.3. Execution and Evaluation Integrity

Execution stops at the first unknown, duplicate, unsupported, colliding, or blocked action. We record legal-prefix fraction, coverage, duplicate status, and the zero-based failure step. An incomplete legal prefix fails success but has no illegal step. Height sorting solves all 10,240 plans in this vertical domain: supporters are lower and already present; an earlier part overlapping the footprint must end below the new part, or the final geometry would collide. Reversing the sequence gives disassembly. This establishes a constraint-following ceiling, not difficult search.

Frozen selection manifests reject unknown/duplicate predictions; missing/malformed submissions remain in applicable denominators. Null means inapplicable, not a removed failure. A versioned strict decoder closes legacy

primitive-type coercion without altering recorded scores. All 117,910 oracle answers pass; independent metrics and mutation checks are detailed in the supplement. Replay labels model, harness, environment, and evaluator operations separately. Visual unfolding of an unordered structure is never called a model plan.

5. Models, Baselines, and Experiments

5.1. Eight-Model Development Matrix

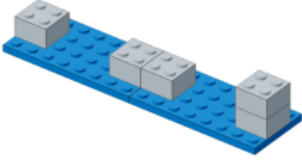
We evaluate the same 39 default/ordinary task cases from three already-exposed development objects (12/20/29 parts) with eight models [2, 7, 21, 23, 27]. Four models retain their original verified responses; four new models receive identical prompts, image bytes, and the 2,200-token JSON output cap. Temperature is zero, model order rotates by case, automatic retries and provider fallback are disabled, and provider, tokens, truncation, latency, and settled OpenRouter charges [24] are retained. Gemini reasoning is disabled; other models have no requested reasoning budget. Cross-round time/provider differences remain potential confounds.

The supplement separates all 13 variants and same-source controls: copying, train-only retrieval, symbolic rules and height sorting. Unsupported pairs are not failures. Table 6 separates format, local changes and exact success; scores do not identify internal perceptual or reasoning mechanisms. In this shared sample, 3 of 13 variants have zero exact success across all eight models and 0 are saturated. Reconstruction successes range from 0/3 to 0/3; mean added-part F1 ranges from 0.0% to 11.1%. These are descriptive ranges, not evidence of a stable population ranking. Floor/ceiling patterns motivate calibrated difficulty and local metrics before a larger comparison.

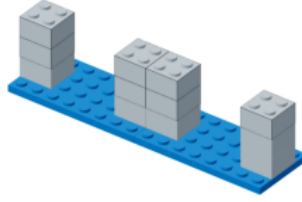
5.2. Paired Tools, Representations, and Training

A preceding 24-source, two-model study contains 674 one-shot calls and 56 paired controls. It is not pooled with the

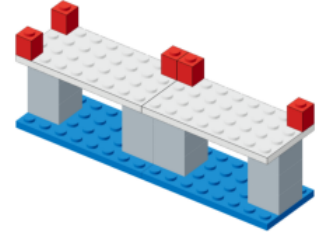
Legal prefix: 7/28 parts



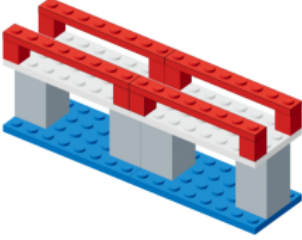
Legal prefix: 14/28 parts



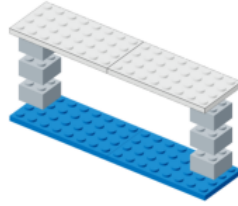
Legal prefix: 21/28 parts



Legal prefix: 28/28 parts



Invalid strategy: decks too early



Blocked inner pier outlined

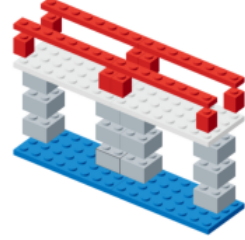


Figure 5. **Ground-truth construction versus a designed dead end.** Four snapshots follow the curated double-span bridge’s verified height-sorted sequence. The last two panels contrast a legal prefix that places both decks before the inner piers with the exploded target; the missing inner pier can no longer descend vertically. The evaluator retains the submitted order and first rejection rather than repairing it.

	Symbolic	RGB	Constraints	RGB	Symbolic	RGB			
Model / algorithm	Relations	Recon.	Gen.	Complete	Edit	Plan	Repair	Format	USD
<i>Vision–language models: same three source objects</i>									
GPT-4.1 mini	3/6	0/3	0/3	0/3	6/9	2/6	4/9	38/39	0.071
GPT-4.1	1/6	0/3	0/3	0/3	6/9	2/6	2/9	35/39	0.341
Gemini 2.5 Flash	2/6	0/3	0/3	0/3	6/9	2/6	3/9	34/39	0.089
Qwen3-VL-32B	0/6	0/3	0/3	0/3	6/9	0/6	1/9	22/39	0.025
Claude Haiku 4.5 ⁺	0/6	0/3	0/3	0/3	6/9	0/6	2/9	22/39	0.317
Gemini 3 Flash (preview) ⁺	3/6	0/3	1/3	0/3	6/9	4/6	1/9	27/39	0.144
Qwen3-VL-8B ⁺	1/6	0/3	0/3	0/3	6/9	0/6	1/9	26/39	0.025
Llama 4 Maverick ⁺	0/6	0/3	0/3	0/3	5/9	1/6	1/9	39/39	0.037
<i>Public-input algorithmic controls: supported tasks only</i>									
Copy current	–	–	–	0/3	0/9	–	3/9	–	–
BOM / extent retrieval	–	0/3	3/3	0/3	–	–	0/9	–	–
Symbolic rules	6/6	–	–	–	9/9	–	–	–	–
Greedy legal action	–	–	–	–	–	5/6	–	–	–
Height-sorted plan	–	–	–	–	–	6/6	–	–	–

Table 5. **Main descriptive matrix: eight models, identical three-source tasks.** Entries are exact successes / cases; variants share sources, so denominators are not independent trials. ⁺ denotes the four newly run models. Algorithm rows use exactly the same cases on supported tasks; – means unsupported or no API charge field, not zero success. No pooled rank or significance claim is supported by three objects. Recognition is separate in the 24-source study because this shared screen has no isolated-part cases.

three-source matrix. Both models fail exact ordinary reconstruction while succeeding on all recolor/removal cases. Eight-source Gemini controls reuse the same first answer for reflection and target-blind validation (Table 7). Symbolic reference supplies privileged geometry; its gain is not a pure perception intervention. A zero exact-success

floor under missing/mismatched images also cannot establish that images are ignored.

Ten SmolVLM-256M [20] base/LoRA [9] pilot jobs share 337 cases and matched supervision, not compute. No positive edit transfer is observed; base outputs all fail formatting. These and earlier Qwen3 [26] pilots are detailed in

Model	Reconstruction		Completion		Planning	Repair	
	Part F1	Surface F1	Added F1	Preserve	Legal prefix	Fault F1	Truncated
GPT-4.1 mini	4.3	18.6	0.0	98.5	49.3	33.3	0/39
GPT-4.1	2.4	18.2	0.0	100.0	51.8	16.7	0/39
Gemini 2.5 Flash	5.8	29.0	11.1	66.7	44.3	27.8	1/39
Qwen3-VL-32B	0.0	0.0	0.0	33.3	1.4	0.0	3/39
Claude Haiku 4.5	1.1	16.1	0.0	100.0	19.7	0.0	2/39
Gemini 3 Flash (preview)	24.7	35.4	11.1	66.7	78.2	16.7	0/39
Qwen3-VL-8B	0.0	0.0	0.0	33.3	0.0	0.0	3/39
Llama 4 Maverick	2.4	15.9	0.0	100.0	39.6	16.7	0/39

Table 6. Task-specific diagnostics on the same three sources, percentages except truncation counts. All returned responses remain in each applicable mean, including format failures. Fault F1 includes only the six faulty repair cases, excluding the three normal controls.

Paired condition	Original	Intervention
no-image	0/8	0/8
mismatched-images	0/8	0/8
symbolic-reference	0/8	8/8
reflection	0/8	0/8
validation	0/8	0/8
repeat	0/8	0/8
relative	6/8	0/8

Table 7. Measured eight-source Gemini controls. Relative denotes assembly representation; other rows are reconstruction. These are a different cohort from Table 5.

the supplement.

5.3. Candidate Consistency and a Visual Control

Coordinate-normalized two-piece probes compare full reconstruction, local pose, candidate selection, and privileged symbolic answers. Candidate lists add information, so they are not a pure output-length change. A 12-source, four-model follow-up explicitly distinguishes reordering from identical repetition. For actual selected poses a_g, a_g^r, a_g^π , the paired endpoint is

$$D_g = \mathbf{1}[a_g^\pi \neq a_g] - \mathbf{1}[a_g^r \neq a_g]. \quad (5)$$

Mean excess mismatch is 0.50–0.75 across models, with all 192 requests retained. Source-bootstrap intervals [3] and per-model counts are in the supplement. Mismatch is inconsistency, not incorrectness, and the same 12 sources do not become 48 independent objects.

A known-renderer mask-matching control uses public BOM and candidates, not target geometry. It solves 16 two-piece sources, preserves selections under permutation, and yields 44/44 image-conditioned successes versus 16 no-image abstentions. This is a signal control, not a general reconstructor.

6. Experimental Coverage and Remaining Work

Diagnostic-2’s public symbolic control passes 36 maintenance cases; a known-renderer color control passes their 36 layer-image counterparts. It reads only public pixels/current geometry, not target JSON, and is not a general pose estimator. The old constant set passes 0/18 new observation cases. Deadlines reject serial schedules in all 18 cases. These checks establish executable signals, not model discrimination. The supplement preserves unmeasured model tables. Independent objects, human review and matched-budget new-model experiments remain necessary.

7. Limitations and Conclusion

BrickAtlas is currently a controlled grid benchmark, not a comprehensive semantic or physically validated assembly benchmark. V2 retains its common base, 19 assembled types, suffix-only completion, topmost repairs, and height-solvable planning. Frontier raises size to 408 parts, but retains the 25-type rectangular vocabulary and has no learned-model measurements. Its access closure is polynomial and local tiling is small; size alone does not prove difficult reasoning or empirical discrimination. The ambiguity extension is exhaustive only within its stated grammar; nine parameter families are not independent external sources. Pixel equality is verified on one renderer, not every camera or human observer. Current human-review packets contain no real submitted judgments. Small adaptive model studies and same-machine replay do not replace independent confirmation and second-environment reproduction.

Within this scope, the benchmark makes useful distinctions executable: changed-region success is not global similarity [6]; an admissible structure does not determine every hidden relation; and a plausible final object does not certify a submitted plan. Stronger claims require the remaining external and human evidence, not more task variants.

References

- [1] Seokjun Ahn, Jungtaek Kim, Minsu Cho, and Jaesik Park. Budget-aware sequential brick assembly with efficient constraint satisfaction. *Transactions on Machine Learning Research*, 2024. 2
- [2] Anthropic. Introducing claude haiku 4.5. <https://www.anthropic.com/news/claude-haiku-4-5>, 2025. 6
- [3] Bradley Efron. Bootstrap methods: Another look at the jack-knife. *The Annals of Statistics*, 7(1):1–26, 1979. 8
- [4] Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. BLINK: Multimodal large language models can see but not perceive. In *European Conference on Computer Vision (ECCV)*, 2024. 2
- [5] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. 2
- [6] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020. 8
- [7] Google DeepMind. Gemini models. <https://ai.google.dev/gemini-api/docs/models>, 2026. Living documentation; run-time identifiers and prices are frozen in BrickAtlas receipts. 6
- [8] Mengqi Guo, Chen Li, Yuyang Zhao, and Gim Hee Lee. TreeSBA: Tree-transformer for self-supervised sequential brick assembly. In *European Conference on Computer Vision (ECCV)*, 2024. 2
- [9] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. 7
- [10] Haochen Huang, Yue Su, Xin Sun, Moonisa Ahsan, Mohammad Aliannejadi, Irene Viola, Zhaochun Ren, Chuang Yu, Aneta Lisowska, Artem Belopolsky, Koen Hindriks, Pablo Cesar, Junxiao Wang, and Jiahuan Pei. LEGO Co-builder: Exploring fine-grained vision-language modeling for multimodal assembly assistants. *arXiv preprint arXiv:2507.05515*, 2025. Version 4 revised July 2026. 2
- [11] Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019. 2
- [12] Chenchen Jing, Yunde Jia, Yuwei Wu, Xinyu Liu, and Qi Wu. Maintaining reasoning consistency in compositional visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. 2
- [13] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and Ross Girshick. CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2901–2910, 2017. 2
- [14] Peter Kulits and Cordelia Schmid. BrickNet: Graph-backed generative brick assembly. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 39252–39261, 2026. 2
- [15] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023. 2
- [16] Jiateng Liu, Bingxuan Li, Zhenhailong Wang, Rushi Wang, Kaiwen Hong, Cheng Qian, Jiayu Liu, Denghui Zhang, Katherine Driggs-Campbell, Manling Li, and Heng Ji. Brick-composer: MLLMs construct everything from building blocks. *arXiv preprint arXiv:2606.05445*, 2026. 2
- [17] Junze Liu, Kun Qian, Florian Dubost, Kai Zhong, Arvind Srinivasan, Nan Chen, Anping Wang, Sam Zhang, Alejandro Mottini, Qingjun Cui, and Tian Wang. 3d primitives are a spatial language for VLMs. *arXiv preprint arXiv:2605.12586*, 2026. 2
- [18] Liang Ma, Jiajun Wen, Min Lin, Rongtao Xu, Xiwen Liang, Bingqian Lin, Jun Ma, Yongxin Wang, Ziming Wei, Haokun Lin, Mingfei Han, Meng Cao, Bokui Chen, Ivan Laptev, and Xiaodan Liang. PhyBlock: A progressive benchmark for physical understanding and planning via 3d block assembly. In *Advances in Neural Information Processing Systems*, 2025. 2
- [19] Wenbo Ma, Daniel Swoboda, Matteo Tschesche, and Till Hofmann. WorkBenchMark: A LEGO-based assembly benchmark with an assembly-by-disassembly baseline for the smart manufacturing league. *arXiv preprint arXiv:2606.19358*, 2026. 2
- [20] Andrés Marafioti, Orr Zohar, Miquel Farré, Merve Noyan, Elie Bakouch, Pedro Cuenca, Cyril Zakka, Loubna Ben Allal, Anton Lozhkov, Nouamane Tazi, Vaibhav Srivastav, Joshua Lochner, Hugo Larcher, Mathieu Morlon, Lewis Tunstall, Leandro von Werra, and Thomas Wolf. SmolVLM: Redefining small and efficient multimodal models. *arXiv preprint arXiv:2504.05299*, 2025. 7
- [21] Meta. The llama 4 herd. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>, 2025. 6
- [22] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Conference on Fairness, Accountability, and Transparency*, 2019. 2
- [23] OpenAI. Introducing GPT-4.1 in the api. <https://openai.com/index/gpt-4-1/>, 2025. Official model announcement; accessed 2026-09-14. 6
- [24] OpenRouter. Api reference. <https://openrouter.ai/docs/api-reference/overview>, 2026. Accessed 2026-09-14; receipts, not current prices, determine reported cost. 6
- [25] Ava Pun, Kangle Deng, Ruixuan Liu, Deva Ramanan, Changliu Liu, and Jun-Yan Zhu. Generating physically stable and buildable brick structures from text. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14798–14809, 2025. 2

- [26] Qwen Team. Qwen3-0.6B model card. <https://huggingface.co/Qwen/Qwen3-0.6B>, 2025. Checkpoint revision c1899de289a04d12100db370d81485cdf75e47ca. 7
- [27] Qwen Team. Qwen3-VL: Official repository and model cards. <https://github.com/QwenLM/Qwen3-VL>, 2025. 6
- [28] Gopal Sharma, Rishabh Goyal, Difan Liu, Evangelos Kalogerakis, and Subhransu Maji. CSGNet: Neural shape parser for constructive solid geometry. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5515–5523, 2018. 2
- [29] Kexian Tang, Junyao Gao, Yanhong Zeng, Haodong Duan, Yanan Sun, Zhening Xing, Wenran Liu, Kaifeng Lyu, and Kai Chen. LEGO-Puzzles: How good are MLLMs at multi-step spatial reasoning? *arXiv preprint arXiv:2503.19990*, 2025. Revised public submission includes a 1–8-step planning set; accessed 2026-09-15. 2
- [30] Aaron Walsman, Muru Zhang, Klemen Kotar, Karthik Desingh, Ali Farhadi, and Dieter Fox. Break and make: Interactive structural understanding using LEGO bricks. In *European Conference on Computer Vision (ECCV)*, 2022. 2
- [31] Ruocheng Wang, Yunzhi Zhang, Jiayuan Mao, Chin-Yi Cheng, and Jiajun Wu. Translating a visual LEGO manual to a machine-executable plan. In *European Conference on Computer Vision (ECCV)*, 2022. 2
- [32] Hongyu Yan and Yadong Mu. Neural assembler: Learning to generate fine-grained robotic assembly instructions from multi-view images. *arXiv preprint arXiv:2404.16423*, 2024. 2
- [33] Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations (ICLR)*, 2024. 2